

EXECUTIVE PRODUCT BLUEPRINT & OPERATIONAL SPECIFICATION

Distributor Retailer Cash Collection & Operations Management Platform

Enterprise hybrid logistics & fintech architecture built to eliminate physical cash collection hazards: Retailer liquidity freezes, untracked field boys, ₹2-3 Lakh cash theft risks, note discrepancies, and manual evening book reconciliation.

MOBILE TECH Flutter (Apps)	BACKEND & DB Laravel/Node • MySQL	GEOFENCE LOCK ≤ 100m PostGIS	APP DEV QUOTE ₹35k + 18% GST	SERVER HOSTING ₹15,000 / Year	AMC & WARRANTY 3 Mo Free • ₹15k/yr
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🔧 Production Stack: Flutter (Collector & Retailer Apps) • MySQL 8.0 • Laravel (PHP) / Node.js API	🛡️ SLA & Support: First 3 Months Free Minor Changes • Post-3 Months: ₹3,000-₹6,000 / change (or ₹15k/yr AMC)
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📊 INFOGRAPHIC: The 4-Step Distributor Cash & Liquidity Closed Loop (कैश चक्र)

ZERO LIQUIDITY TRAP

🔗 Page 6: Real Workflow →

STEP 1: LOAD (पैसा डाला)

Digital Disbursal

Distributor credits Retailer wallet ₹50k-₹1L for AEPS/DMT.

STEP 2: CASH-OUT (कैश आया)

Customer AEPS

Villagers withdraw cash; Retailer drawer fills with physical cash.

STEP 3: PICKUP (कैश उठाया)

1-Tap App Settle

Collector ≤100m arrives; 1-Tap Approval loads wallet for Recharge/BBPS!

STEP 4: VAULT (बैंक में जमा)

Bank Deposit

Collector deposits cash at Hub; Bag float resets to ₹0 & banked.

🔥 Real-World Distributor Cash Collection Problems & System Solutions

Root Cause Operational Analysis

PROBLEM #1: RETAILER WALLET LOAD VS CASH RECOVERY

Wallet Me Balance Dala, Counter Par Cash Aaya

Distributor retailers ke wallet me digital amount dalta hai. Retailer AEPS cash withdraw karke cash counter par ikattha kar leta hai aur wallet balance khatam ho jata hai. Distributor ko pata nahi chalta kiski kitni request aayi aur kitna cash physical uth gaya.

PROBLEM #2: FIELD BOYS PAR ZERO VISIBILITY

Untracked Movement & False Excuses

Distributors have zero insight into where field boys are traveling, which shops they visited, or whether claims of "Shop was closed" or "Retailer was out" are genuine or deliberate delays.

SOLUTION #1: WALLET VS CASH RECONCILER + BBPS SERVICES

Live Request Ledger + 1-Tap Approval & Utility Suite

Admin screen par live dikhta hai: kis retailer ki kitni request aayi aur kitni collection physical uth gayi. Collector dwara physical cash verify karke [APPROVE] tap karte hi retailer wallet turant recharge, electricity bill payment aur AEPS ke liye reload ho jata hai (Zero OTP delay).

SOLUTION #2: LIVE GPS TELEMETRY & 100M GEOFENCE

Continuous GPS Tracking + Hardware Lock

The Admin Live Map tracks moving bike markers every 10 seconds. The mobile app's "Receive Cash" button is **strictly geofence-locked** and only activates when the collector is within 100m of the shop.

🔑 Official System Login Credentials & Role Directory (Test Accounts)				Single Sign-On (SSO) & Device Binding Active
SYSTEM ROLE	USER ID / USERNAME	MOBILE NUMBER	DEFAULT PASSWORD	PORTAL SCOPE & LEVEL
SUPER ADMIN	admin@rockvanta.com	+91 98120 00001	Admin@2026#CMS	Central Live Map, Auto-Dispatch Engine, Disputes & Vault
DISTRIBUTOR	dist.rohtak@rockvanta.com	+91 98120 00002	Dist@Rohtak#2026	Sub-Network Retailers, Vault Handover Desk, Float Reset
CASH COLLECTOR	col.rahul104	+91 98120 00003	Collector@Rahul#104	Flutter Mobile App, 100m Geofence, Float Meter, Thermal Print
RETAILER	ret.abcdigital	+91 98120 00004	Retailer@ABC#88	Merchant Web/App, Request Cash Pickup, AEPS, DMT Wallet

2. Admin & Distributor Web Control Panel (एडमिन / डिस्ट्रीब्यूटर वेब पोर्टल के सारे फीचर्स)

360° Financial Ledger, Reconciler Desk, Full CRUD & Fleet Operations

🏠 A. 360° Financial Khata & Real-Time Cash Audit Desk (उधार, रिसीव्ड व दिया गया कैश बहीखाता)

ADMIN 360° LIVE LEDGER

1. CASH RECEIVED (आया)
₹4,85,000
18 Retailer Pickups Today

2. WALLET CREDITED (दिया)
₹4,85,000
Instant Digital Settlement

3. RETAILER UDHAR (उधार)
₹2,40,000
Total Disbursed Market Float

4. CASH IN TRANSIT (बैग में)
₹1,12,000
Held by 4 Field Collectors

2. RETAILER / SHOP DETAILS	WALLET DISBURSED	REQUESTS RAISED (आई)	CASH COLLECTED (हुई)	PENDING (बाकी)	LIVE STATUS
Radhe Digital Store (+91 98120 00004 • Ward 4)	₹1,00,000	2 Reqs (₹70,000)	₹50,000 (By Rahul)	₹20,000	🟡 En Route
Sharma Telecom & AEPS (+91 98120 00012 • Station Rd)	₹60,000	1 Req (₹40,000)	₹40,000 (By Amit)	₹0	🟢 100% Settled
Gupta Grahak Seva Kendra (+91 98120 00015 • Main Chowk)	₹85,000	3 Reqs (₹85,000)	₹60,000 (By Rahul)	₹25,000	🔴 Assigned (Timer)

3. 📅 Date-Wise History: Admin calendar picker se kisi bhi date ka total cash collection record dekha ja sakta hai.

25-Sep: ₹4.85L (18 Pickups)

24-Sep: ₹5.12L

4. 🚨 Daily Outstanding (1/Day Alert)

5. 📄 Tally/Excel Export

📶 VISUAL RADAR: Live Moving Fleet GPS Map Telemetry (Rohtak Urban Zone)

4 ONLINE • 0 SOS ALERTS MAPBOX VCTR • 10s PING

Rahul (COL-104)
Loc: Ward 4 • Batt: 84%
Holding: ₹40,000 / ₹1L

Amit (COL-102) 0 KM/H (At Shop)
Loc: Station Rd • Batt: 92%
Holding: ₹40,000 / ₹1L

Sonu (COL-108)
Loc: Main Chowk • Batt: 68%
Holding: ₹32,000 / ₹1L

Central Vault Desk
Evening Closing Station
Vault: ₹4,85,000 Locked

HUB ACTIVE

🚦 B. Live Operations Radar & GPS Tracking Map

Real-time live fleet tracking across polygon delivery zones with emergency SOS red alerts.

6. Real-Time Moving Fleet Map: Sabhi active collectors ki bike speed, live location, battery % aur bag mein kitna cash hai, sab map par live dikhta hai.

7. Polygon Territory Zones: Pura shahar/kasba wards aur zones mein divide hota hai (Ward 1, Ward 2, Rural East, etc.).

8. SOS Central Red Alert Screen: Agar kisi collector ne emergency button dabaya, to Admin dashboard par siren bajega aur police/supervisor alert trigger ho jayega.

📄 D. Bulk Mobile Excel Upload & File Attachment

Bulk onboarding of hundreds of retailers via Excel sheet mapping to verified mobile numbers.

13. Bulk Excel/CSV Import: 100+ retailers ka data (Mobile No, Shop Name, GPS, assigned area) ek Excel sheet upload karke second-o me onboard karna.

14. Document Attachment Engine: Retailer aur collector ke mobile number ke sath unke GST certificate, shop photo ya aadhar card files attach karna.

14b. Admin Flash Pop-up & Broadcast Notice Engine: Admin 1-click me sabhi retailers ko custom pop-up message bhej sakta hai (Bank holiday, cash timing, notices) jo screen par live flash modal ban kar show hota hai.

14c. Daily Outstanding Reminder Engine (दिन में 1 बार बकाया अलर्ट): जिन रिटेलर्स की उधारी/बकाया पेंडिंग है, Admin 1-क्लिक से या रोज़ाना सुबह 9:00 AM ऑटोमैटिक क्रॉन से उन्हें नोटिफिकेशन भेज सकता है (24-Hour Anti-Spam Lock).

🏢 C. Complete Admin CRUD & Multi-Bank Management

Full Create, Read, Update, Delete & Multi-Account Deposit tracking across entities.

9. Retailers CRUD: New shop add karna, GPS pin karna, default collector assign karna, penalty rules set karna, aur shop block/activate karna.

10. Collectors CRUD: Field boys add karna, bike number, mobile number, float limit (₹1,00,000) set karna aur duty suspend karna.

11. Pickup Requests CRUD: Live incoming requests dekhna, kisi collector ko manually re-assign karna, ya request cancel karna.

12. Multi-Bank Accounts CRUD & Deposit Tracker: Admin apne multiple company bank accounts add kar sakta hai (HDFC, ICICI, SBI) aur track kar sakta hai kis bank account me kitna cash deposit hua (Deposit Slips, UTR, Bank Balance).

🔒 E. Evening Vault Handover Desk & Auto-Penalty Rules

Dedicated day-end closing station, bank deposit allocation, note verification, and penalty rules.

15. Evening Vault & Bank Deposit Allocation: Sham ko physical cash note counting machine se verify hota hai; Admin choose karta hai kaun se bank account me kitna deposit hoga (e.g. ₹3L HDFC + ₹1.85L SBI) aur Digital Vault Sign-off se collector float turant ₹0 reset karta hai.

16. Auto-Penalty Configuration: Retailer cancellation penalty slab set karna (₹50-₹200). Post-dispatch/100m cancel par ₹50 auto-deduct (₹35 collector petrol + ₹15 admin).

17. Admin Penalty Waiver: Emergency mamlo me Admin 1-click se penalty refund/waive kar sakta hai.

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Page 2 of 6

1. Cash Collector Mobile App (कलेक्टर मोबाइल ऐप के सारे फीचर्स)

Fast Fleet Acceptance, 100m Geofenced Unlock, Float Meter & Silent SOS Shield

A. LOGIN, ATTENDANCE & SECURITY GUARD

DEVICE BINDING • FLOAT METER

- ✓ **1. Device Binding (IMEI Lock):** Collector sirf apne registered official phone se login kar sakta hai. Doosre phone se login allow nahi hoga.
- ✓ **2. Geofenced Duty Punch-In / Out:** Duty start karte waqt collector ka live GPS location aur bike ka initial odometer (KM) log hota hai.
- ✓ **3. Live Float Meter (कैश बैग मीटर):** App ke top par real-time progress bar chalti hai (e.g. ₹82,500 / ₹1,00,000 Limit). Agar collector ke bag mein ₹1,00,000 cash ho jata hai, to nayi jobs auto-lock ho jati hain jab tak wo vault mein cash jama na kare.

B. BROADCAST ALERT & JOB ACCEPTANCE

SIMULTANEOUS PUSH • LIVE COUNTDOWN

- ✓ **4. Simultaneous Fleet Broadcast:** Retailer ki request aate hi territory ke sabhi active online collectors ko ek sath loud alert sound ke sath push notification aati hai.
- ✓ **5. Persistent Reverse Countdown Timer:** Screen par ticking countdown timer chalta hai (90s... 89s... 88s...).
- ✓ **6. First-to-Accept Race Lock:** Jo collector sabse pehle [ACCEPT & LOCK] tap karega, job uske naam par book ho jayegi aur baaki sabhi collectors ke phone se dismiss ho jayegi.
- ✓ **7. Instant Identity Broadcast:** Accept karte hi collector ka **Name (Rahul Kumar)**, **Mobile No** aur **ID** Retailer aur Admin screen par show ho jata hai.

LIVE BROADCAST (SENT TO ALL ONLINE COLLECTORS)

01:18 REMAINING

Shop: Radhe Digital Store • Ward: Ward 4, Main Road

Cash: ₹40,000 (80 x ₹500) • Dist: 1.2 KM • **First to Accept Wins!**

✕ Pass

✓ ACCEPT & LOCK

✓ CLAIMED BY: Rahul Kumar (ID: COL-104 • Mob: +91 98120 00003)

SHOWN TO RETAILER & ADMIN

● **Rahul (Collector 1)**

Tapped [ACCEPT] @ 01:18s → ✓ LOCKED

● **Amit (Collector 2)**

Tapped 1.2s Later → ✕ DISMISSED

● **Sonu (Collector 3)**

Missed Countdown → ✕ DISMISSED

C. ROUTE NAVIGATION & D. ZERO-PHOTO 100M GEOFENCED COLLECTION DESK

100% PHOTO-FREE • ZERO-OTP

- **8. In-App Navigation:** Dukaan tak pahunchne ke liye Google Maps direction milti hai.
- **9. 1-Tap Masked Call:** Collector bina phone number save kiye seedha retailer ko call karta hai.
- **10. 100% Photo-Free (Zero Camera):** Koi selfie ya photo lene ka jhanjhat nahi.
- **11. 100m Hardware Geofence Unlock:** "Receive Cash" button dukaan ke 100m daayre me physically aane par hi unlock hota hai.
- **12. Smart Note Denomination Desk (कैश नोट कैलकुलेटर):** App me full note breakdown keypad milta hai: ₹500 x 70 = ₹35k, ₹200 x 20 = ₹4k, ₹100 x 10 = ₹1k → **Total ₹40,000 (81 Notes).**
- **13. Partial Cash Collection:** Graahak aane par collector amount modify kar sakta hai (₹50k → ₹30k) with retailer confirm.
- **14. Wallet Loading & Collector Approval:** Note count match hote hi collector amount add karke [ADD TO WALLET & APPROVE] tap karta hai (No OTP).

IN-APP DESK: Cash Denomination Calculator & Currency Note Breakdown (Zero-OTP)

GPS: 28m • GEOFENCE UNLOCKED

NOTE DENOMINATION

₹500 x 70 = ₹35k

₹100 x 10 = ₹1k

₹200 x 20 = ₹4k

Other = ₹0

81 NOTES

CALCULATED TOTAL

₹40,000

✓ ADD WALLET & APPROVE

RECEIPT & BAG

PRINT READY

Bag: ₹82,500 / ₹1L

E. RECEIPTS, HANDOVER & EMERGENCY SHIELD

SILENT SOS • THERMAL PRINT • EOD VAULT

- ✓ **16. Bluetooth Thermal Receipt:** Mini pocket printer se surant physical receipt print hoti hai + retailer ko automated WhatsApp PDF receipt chali jati hai.
- ✓ **17. Silent SOS / Panic Shield (लुट व हादसे से सुरक्षा):** Phone ka power button 3 baar dabane par Admin aur nearby collectors ko bina aawaz ke live GPS stream chali jati hai.
- ✓ **18. Evening Handover Mode (ईवनिंग क्लोजिंग):** Sham ko distributor vault par total physical cash machine se ginwa kar jama karta hai aur float wapas ₹0 karta hai.

3. Retailer Portal / Mobile App (रिटेलर ऐप व पोर्टल के सारे फीचर्स)

On-Demand Pickup Request, Collector Tracking, Retailer [Accept & Confirm] & Instant Wallet Balance

A. CASH PICKUP REQUEST & TEAM STAFF MANAGEMENT

SHOP TEAM • CASH DISPATCH

✓ 1. 1-Click On-Demand Request (Anti-Duplicate Lock):

Retailer ko sirf amount daalna hota hai (e.g. ₹40,000) aur request button dabana hota hai. *Critical Rule:* Jab tak previous collection cycle Retailer dwara [ACCEPT] hokar complete nahi hoti, tab tak nayi request send karna locked rehta hai.

✓ 2. Auto Shop GPS Tag:

Request ke sath dukaan ki exact location auto-tag ho jati hai taaki collector seedha dukaan par pahunchhe.

✓ 3. Retailer Team & Operator Staff Add:

Retailer apni dukaan ke liye counter boys ya cash operators add kar sakta hai jo uski taraf se counter par cash handover aur accept verify kar sakein.

B. LIVE TRACKING & COLLECTOR IDENTITY TRANSPARENCY

✓ 4. Collector Identity Screen (कलेक्टर का नाम और फोन आगे दिखेगा):

Request accept hote hi screen par aage show hoga: **Collector:** Rahul Kumar • **Mobile:** +91 98120 00003 (Direct Call) • **ID:** COL-104.

✓ 5. Live Map & Distance Counter:

Retailer dekh sakta hai ki collector bike par kitni door hai (500m door • 2 min mein pahunchne wala hai).

C. COLLECTOR WALLET ADD & RETAILER [ACCEPT & CONFIRM] DESK

ZERO-OTP • ACCEPT TO UNLOCK

✓ 6. Collector Wallet Add & Approval Dispatch:

Dukaan par collector physically cash note ginta hai aur retailer ke wallet me amount add karke [ADD WALLET & APPROVE] tap karta hai (No OTP delay). Isse Retailer phone par live alert trigger hota hai.

✓ 7. Retailer [ACCEPT & CONFIRM] Rule & Request Lockout (जब तक एक्सेट नहीं करेगा, नई रिक्वेस्ट लॉक):

Collector ke approve karte hi retailer screen par [ACCEPT & CONFIRM] button aayega. Retailer jab tak accept nahi karega, tab tak nayi cash request bhejna permanently locked rahega. Retailer ke accept karte hi 1-sec me ₹40,000 credit finalize hota hai aur agla request desk unlock hota hai.

✓ 8. Mobile Recharge & Bill Payment Suite:

Reloaded balance se retailer turant all-operator mobile recharge (Jio, Airtel, Vi, BSNL) aur Utility Bill Payments (Bijli, Pani, Gas, DTH, FASTag) shuru kar sakta hai.

D. RULES, CANCELLATION, 1-TIME REMINDER & POP-UPS

✓ 9. 3-Minute Free Cancel Window:

Request daalne ke baad 3 minute tak free cancel kar sakta hai; uske baad ₹50 fuel charge wallet se deduct hoga.

✓ 10. Real-time Passbook & Khata:

Kis date ko kitna cash collector ko diya, kitna wallet recharge me use hua, sabka date-wise statement.

✓ 11. Instant WhatsApp Receipt:

Har transaction ki digital PDF receipt retailer ke WhatsApp par auto-deliver ho jati hai.

✓ 12. Daily 1-Time Outstanding Reminder (दिन में 1 बार बकाया अलर्ट):

जिन रिटेलर्स की उधारी पेंडिंग है, उन्हें रोज़ाना सुबह 9:00 AM सिस्टम पुश नोटिफिकेशन व ऐप कार्ड में दिन का बकाया दिखाता है (Strict 24-Hour Anti-Spam Lock).

✓ 13. Admin Live Flash Pop-up Alert:

Admin dwara emergency notice (Bank holiday, cash timings) sabhi retailers ke screen par live modal popup bankar show hota hai.

RETAILER SCREEN: Collector Added to Wallet & Approved → Retailer [ACCEPT & CONFIRM] Desk

ZERO-OTP • ACCEPT TO UNLOCK

ADMIN POP-UP: "Bank Holiday Pickups till 8 PM"

OK

DAILY OUTSTANDING: ₹14,500 (1 Notice/Day)

PAY

ASSIGNED COLLECTOR
Rahul Kumar (COL-104)
+91 98120 00003 [Call]

COLLECTOR ADDED & APPROVED: ₹40,000

✓ [ACCEPT & CONFIRM]

Tap to Finalize Credit & Unlock New Request

NEW REQUEST: LOCKED

Unlocks after Accept

Recharge BBPS Bill

END-TO-END RETAILER OPERATIONAL FLOW (दुकानदार का शुरू से अंत तक का 5-चरणीय चक्र)

100% COMPLETE MERCHANT LIFECYCLE

1. RAISE REQ
• Enter ₹40,000
• Auto GPS Tag
• 90s Broadcast Race

→

2. RADAR & CALL
• Rahul (COL-104)
• 500m Radar
• Direct Call

→

3. CASH & NOTE COUNT
• At Shop (≤100m)
• 81 Notes Count
• [Add & Approve]

→

4. [ACCEPT] UNFREEZE
• [ACCEPT] Tap
• Next Request LOCKED until Accepted

→

5. RECHARGE & UTILITY
• +₹40k Loaded
• Mobile & BBPS
• Desk Unlocked

4-Stage Operational Flowchart & 3-Way Unlocked Outcomes:

STRICT RETAILER LOCKOUT RULE

[1. Retailer Request] → [2. Collector Accepts] → [3. Collector Adds to Wallet & Approves] → [4. Retailer Taps [ACCEPT & CONFIRM]]

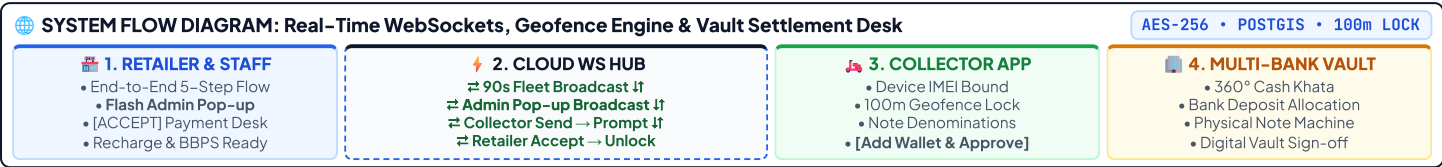
Wallet Balance Final Credit (+₹40,000 Loaded in 1 Sec)

Mobile Recharge & BBPS Utility Unlocked (Jio, Airtel, Electricity, Water)

"New Cash Pickup Request" Desk UNLOCKED! (Next Cycle Active)

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Page 4 of 6



🕒 Real-World End-to-End Daily Workflow (सुबह 8:30 AM से रात 9:30 PM तक)

Chronological Ground Reality: Retailer Cash Handover, Collector Note Counting & Evening Bank Vault Handover

STEP 1
Retailer Req
₹40k Pickup Send

STEP 2
90s Siren Race
Collector [ACCEPT]

STEP 3
≤100m Arrival
PostGIS Unlock

STEP 4
81 Notes Count
500x70 + 200x20...

STEP 5
Col. Approve
Zero-OTP Thermal Slip

STEP 6
Ret. [ACCEPT]
Wallet + BBPS Ready

STEP 7
Vault & Banks
HDFC + SBI Recon

🕒 08:30 AM — Phase 1: Morning Duty Start, Security Guard & Daily Outstanding Reminder

IMEI LOCK • DAILY 1-TIME ALERT

Collector Rahul Kumar designated ward (Rohtak Urban) me duty start karta hai → Hardware IMEI & Anti-Mock GPS pass hota hai → Bike meter log karke punch-in karta hai (Float: ₹0 / ₹1,00,000). Saath hi Automated Daily Outstanding Cron (दिन में 1 बार बकाया अलर्ट) un sabhi retailers ko unke pending khat ke notification bhej deta hai.

🕒 10:15 AM — Phase 2: Retailer Cash Buildup & 90-Second Broadcast Race

90s RACE • FIRST-ACCEPT WINS

DMT/AEPS cash withdrawal se retailer counter par ₹40,000 physical cash jama ho jata hai. Retailer 1-click se ₹40,000 Pickup Request bhejta hai. Dukaan ki GPS location ke sath 90s ka broadcast alert nearby collectors ko jata hai. Rahul siren bajte hi 14s me [ACCEPT & LOCK] tap karta hai. Retailer screen par Rahul ka Naam, Mobile (+91 98120 00003), aur Bike Radar (500m • 2 Min) live show ho jata hai.

🕒 10:24 AM — Phase 3: ≤100m Arrival, Physical Cash & Smart Denomination Desk

≤100m UNLOCK • 81 NOTES

Rahul dukaan ke samne bike park karta hai (GPS distance 28m ≤ 100m → "Receive Cash" button unlock). Retailer cash deta hai. Rahul Smart Denomination Calculator me notes feed karta hai: ₹500 × 70 = ₹35k, ₹200 × 20 = ₹4k, ₹100 × 10 = ₹1k → Total ₹40,000 (81 Notes). Note count match hote hi Rahul [ADD WALLET & APPROVE] tap karta hai (Zero-OTP). Pocket printer se 2 slips print hoti hain.

🕒 10:28 AM — Phase 4: Retailer [ACCEPT & CONFIRM], Request Unfreeze & BBPS Active

ACCEPT TO UNLOCK • RECHARGE READY

Retailer phone par live alert bajta hai: [🔔 Collector Submitted ₹40,000]. Strict Rule: Jab tak Retailer [ACCEPT & CONFIRM] nahi karega, tab tak agli pickup request create karna permanently locked rahega! Retailer [ACCEPT] tap karta hai → 1-second me ₹40,000 wallet load confirm hota hai → Retailer turant Jio/Airtel Recharge aur Bijli/Pani Bill payment shuru karta hai aur new request desk unlock ho jata hai.

🕒 07:30 PM — Phase 5: Evening Vault Handover, Multi-Bank Allocation & 360° EOD Audit

NOTE MACHINE • MULTI-BANK RECON

Din bhar me 7 shops se collection ke baad Rahul ka bag ₹94,500 par pahunchta hai (near ₹1L safety cap). Rahul Central Vault Desk par cash machine me note ginwata hai. Admin cash ko multi-bank accounts me allocate karta hai: ₹50,000 HDFC Current Account + ₹44,500 SBI Current Account. Admin Digital Vault Sign-off enter karta hai → Rahul ka app float wapas ₹0.00 ho jata hai. Duty punch-out ke baad Admin 360° Master Reconciler se Zero-Difference Ledger close karta hai.

👥 REAL-WORLD 3-ROLE OPERATIONAL SWIMLANE (जमीनी 3-तरफा भूमिका व समन्वय)

ZERO-DISCREPANCY WORKFLOW

🏪 RETAILER LANE

1. ₹40k Cash Pickup Request Bhejna
2. Collector Radar & Identity Check
3. Counter Par Physical Cash Handover
4. [ACCEPT & CONFIRM] Tap Karna
5. Wallet Se Recharge & Bill Payment

🚒 CASH COLLECTOR LANE

1. IMEI Geofenced Punch-in (₹0 Float)
2. 90s Race Alert Me [ACCEPT & LOCK]
3. ≤100m Arrival & 81 Notes Count
4. [ADD WALLET & APPROVE] Tap
5. Evening Vault Deposit (HDFC/SBI)

🏦 ADMIN / VAULT LANE

1. Live Bike Speed, Batt & Cash Radar
2. Flash Pop-up Notice Broadcast
3. Multi-Bank Accounts Management
4. Note Counting Machine Cash Verify
5. 360° Reconciler Ledger Close

🔒 Ground Reality Edge-Case Protocols (जमीनी व्यावहारिक समस्याओं के समाधान):

• 1. Cash Kam Padna (Partial Collection): Agar graahak aane se cash ₹40k se ₹25k ho gaya, to collector amount modify karta hai aur retailer phone par 1-tap confirm pop-up se settle hoti hai.
• 2. 100m Geofence Protection: Collector dukaan se 100m se door reh kar payment submit nahi kar sakta; PostGIS check hardware level par lock rakhta hai.

• 3. Basement No-Network (Offline Token): Basement shops me internet na hone par app offline cryptographic hash banata hai jo collector bina net ke verify kar sakta hai.
• 4. ₹1 Lakh Safety Hard Lock: Collector bag me ₹1 Lakh hote hi app new assignments block kar deta hai jab tak distributor vault me cash jama na ho.